Probability and Statistics – BSDSF23 (Afternoon)

Total Time: 50 minutes **Quiz 4** **Maximum Marks: 50**

Instructor: Dr. Faisal Bukhari **Obtained Marks:__________**

Name: ____________________________________ **Roll No. ___________________**

1.	The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, automobile, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by automobile, and 2 arrived by train?	**(10)**

$n = 9$ **(total number of delegates)**
$x_1 = 3$ **(arrived by air)**
$x_2 = 3$ **(arrived by bus)**
$x_3 = 1$ **(arrived by automobile)**
$p_1 = 0.4$ **(probability of arriving by air)**
$p_2 = 0.2$ **(probability of arriving by bus)**
$p_3 = 0.3$ **(probability of arriving by automobile)**
$p_4 = 0.1$ **(probability of arriving by train)**

Using Multinomial Distribution:

$$P(X) = \left(\frac{9!}{(3!\times 3!\times 1!\times 2!)}\right) \times (0.4)^3 \times (0.2)^3 \times (0.3)^1 \times (0.1)^2$$
$$= 0.0077$$

2.	An automobile manufacturer is concerned about a fault in the braking mechanism of a particular model. The fault can, on rare occasions, cause a catastrophe at high speed. The distribution of the number of cars per year that will experience the catastrophe is a Poisson random variable with $\lambda = 5$.	**(15)**

a) What is the probability that at most 3 cars per year will experience a catastrophe?

λ = 5 per year
Using Poisson Distribution:
$P(X \leq 3) = 0.2650$

b) What is the probability that more than 1 car per year will experience a catastrophe?

$P(X > 1) = 1 - 0.0404$
$= 0.9596$

3.	According to a study published by a group of University of Massachusetts sociologists, about two thirds of the 20 million persons in this country who take Valium are women. Assuming this figure to be a valid estimate, find the probability that on a given day the fifth prescription written by a doctor for Valium is	**(15)**

a) the first prescribing Valium for a woman

$p = \frac{2}{3}$

$g(x; p) = p\, q^{x-1}$

$g\left(5; \frac{2}{3}\right) = \frac{2}{3}\left(\frac{1}{3}\right)^4$

$= \frac{2}{243}$

b) the third prescribing Valium for a woman.
$p = \frac{2}{3}$ $k = 3$ **Using the negative binomial distribution,** $b^*(x;\ k, p) = \ _{x-1}C_{k-1}\ p^k q^{x-k}$ $b^*\left(5;\ 3, \frac{2}{3}\right) = \ _4C_2 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2$ $= \frac{16}{81}.$
4. A foreign student club lists as its members 2 Canadians, 3 Japanese, 5 Italians, and 2 Germans. If a committee of 4 is selected at random, find the probability that all nationalities except Italian are represented. **(10)**
The extension of the hypergeometric distribution gives a probability $$\frac{\binom{2}{1}\binom{3}{1}\binom{2}{2}}{\binom{12}{4}} + \frac{\binom{2}{2}\binom{3}{1}\binom{2}{1}}{\binom{12}{4}} + \frac{\binom{2}{1}\binom{3}{2}\binom{2}{1}}{\binom{12}{4}} = \frac{8}{165}$$